

Climate Risk & Health Prediction Challenge

Binary classification of climate-sensitive mortality — Zindi competition, 2026

1. Problem Statement

The challenge asks for a binary classifier predicting whether a recorded death in Uganda falls into a “climate-sensitive” category, using demographic (age, gender, zone), geographic (location, elevation, slope), and climate (temperature, rainfall, NDVI) features. Submissions are scored on a weighted blend of F1 (60%) and ROC-AUC (40%), with the classification threshold fixed at 0.5 per the competition rules.

2. Dataset

- Train.csv / Test.csv: demographic and base climate fields (zone, gender, age, temperature, precipitation, latitude/longitude, location, deathdate) — 3,146 / 1,030 records.
- climate_features.csv: 17 engineered climate/terrain columns (7/30/90-day rainfall and temperature windows, NDVI, elevation, slope), keyed by ID, covering both Train and Test.
- Target distribution: 65% climate-sensitive / 35% not.
- location is high-cardinality (dozens of distinct villages) relative to the dataset's size, which shaped the encoding strategy described below.

3. Methodology — Iteration Timeline

Seven notebooks trace a steady, score-driven progression from a minimal baseline (0.793) to the final feature-engineered CatBoost pipeline (0.84562). Each iteration's leaderboard score is the one recorded at the top of that notebook:

Notebook	What changed	Leaderboard score
baseline.ipynb	Logistic Regression only; location parsed into split string columns; no external climate data	0.793
feature-engineered.ipynb	Broadened to 5 candidate models (Logistic Regression, Random Forest, Extra Trees, Gradient Boosting, XGBoost)	0.818
fe-hpt.ipynb	Hyperparameter search (RandomizedSearchCV/GridSearchCV) across all tree-based models	0.836
hpt-eda-updated.ipynb	Correlation-heatmap review of engineered features; refined encoding based on which location values carried signal	0.8362
fe-updated.ipynb	Added interaction features (temperature x precipitation, geographic x demographic terms)	0.83908
fe-catboost.ipynb	Introduced CatBoost as an additional model in the comparison	0.84463
catboost-climate-features.ipynb	Merged climate_features.csv (7/30/90-day rainfall & temperature windows, NDVI, elevation, slope); target-mean encoding retained across all location values	0.84562 (best)

4. Key Design Decisions

4.1 Target-mean encoding for high-cardinality categoricals

location, zone, gender, and age_group were encoded using target-mean encoding rather than one-hot encoding. With location alone spanning several dozen distinct values on a dataset of only ~3,100 rows, one-hot encoding would have produced a very sparse, high-dimensional feature space relative to the available data — target-mean encoding compresses each category to a single informative number instead. All location values were retained rather than bucketing or dropping low-frequency ones, since with a dataset this small, every additional data point per category carries meaningful weight; dropping rare locations would have discarded information the model could otherwise use.

4.2 Progressive model comparison

The model set was widened iteration by iteration — from a single Logistic Regression baseline to a six-model comparison (Gradient Boosting, Extra Trees, Random Forest, XGBoost, LightGBM, CatBoost) — with the strongest performer selected at each stage based on its train/test score consistency.

4.3 Feature engineering additions

- Interaction terms: temperature x precipitation, geographic x demographic combinations.
- External climate data merge: 7/30/90-day rainfall and temperature windows, NDVI, elevation, and slope from climate_features.csv, added in the final iteration.
- Correlation-driven review: an intermediate pass examined the correlation heatmap of engineered features to check which encoded location values carried genuine signal before finalizing the encoding approach.

5. Tools & Techniques

- Languages / libraries: Python, pandas, NumPy, scikit-learn, XGBoost, LightGBM, CatBoost
- Modeling: Logistic Regression, Random Forest, Extra Trees, Gradient Boosting, XGBoost, LightGBM, CatBoost
- Encoding: target-mean encoding for high-cardinality categorical variables (location, zone, gender, age group)
- Feature engineering: interaction terms, external climate-window features (rainfall/temperature over 7/30/90 days, NDVI, terrain)
- Hyperparameter tuning: RandomizedSearchCV / GridSearchCV across tree-based models

6. Results

The pipeline improved from 0.793 to 0.84562 on the public leaderboard across seven iterations, driven by (in order of impact): introducing CatBoost, merging external climate-window features, adding interaction terms, hyperparameter tuning, and widening the model comparison beyond a single linear baseline.

7. Key Takeaways

- Iteratively improved a binary classifier's leaderboard score from 0.793 to 0.846 through systematic feature engineering and model selection.
- Applied target-mean encoding to handle a high-cardinality categorical variable (location) on a small dataset, preserving information from low-frequency categories rather than discarding it.
- Compared seven model families (linear and tree-based/gradient-boosted) and selected the best performer using train/test score consistency at each stage.
- Integrated external climate-window features (multi-horizon rainfall, temperature, NDVI, terrain) to lift the final model beyond what demographic and single-point climate readings alone could achieve.